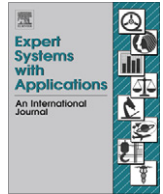




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Incorporating label dependency into the binary relevance framework for multi-label classification

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ABSTRACT

In multi-label classification, examples can be associated with multiple labels simultaneously. The task of learning from multi-label data can be addressed by methods that transform the multi-label classification problem into several single-label classification problems. The binary relevance approach is one of these methods, where the multi-label learning task is decomposed into several independent binary classification problems, one for each label in the set of labels, and the final labels for each example are determined by aggregating the predictions from all binary classifiers. However, this approach fails to consider any dependency among the labels. Aiming to accurately predict label combinations, in this paper we propose a simple approach that enables the binary classifiers to discover existing label dependency by themselves. An experimental study using decision trees, a kernel method as well as Naïve Bayes as base-learning techniques shows the potential of the proposed approach to improve the multi-label classification performance.

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1. Introduction

In traditional supervised learning, each example in the training set is associated to a single-label y_j from a set of disjoint labels L , i.e., $y_j \in L$, where $L = \{y_j: j = 1 \dots q\}$ is the set of q possible labels. Thus, the single-label classification task deals with classifying examples into a single label. On the other hand, in multi-label learning, each example in the training set is associated with multiple labels simultaneously, and the multi-label classification task deals with classifying each example into a subset of labels $Y_i \subseteq L$.

Multi-label learning is an emerging and promising research topic of machine learning due to the increasing number of new applications where examples are annotated with more than one label. Multi-label classification is being used in applications such as semantic annotation of video (Qi et al., 2007) and image (Yang, Kim, & Ro, 2007), bioinformatics (Blockeel, Schietgat, Struyf, Džeroski, & Clare, 2006; Vallim, Duque, Goldberg, & Carvalho, 2009), text categorization (Chang, Cheng, & Liau, 2008), music categorization into emotions (Trohidis, Tsoumakas, Kalliris, & Vlahavas, 2008) and automatic image annotation (Feng & Xu, 2010), to name just a few.

According to Tsoumakas, Katakis, and Vlahavas (2009), where a good survey on multi-label classification is presented, methods for multi-label classification can be grouped into two main categories:

problem transformation and algorithm adaptation. The first group of methods is algorithm independent and they transform the multi-label classification problem into either one or more single-label classification problems. The second group of methods extends specific learning algorithms in order to handle multi-label data directly. Label Powerset (LP)¹ and Binary Relevance (BR) are two well-known problem transformation methods.

The basis of LP method is to combine entire label sets into single labels to form a single-label problem such that the set of single labels represents all distinct label subsets in the original multi-label representation. In other words, LP considers each different k set of labels Y_i as one of the classes of a new single-label classification task. Although LP takes into account label dependency, a drawback is the large number of classes that LP has to deal with when the number of a different set of labels k is large, which makes the learning process difficult.

In the Binary Relevance approach transformation method, a multi-label classification problem is decomposed into a binary classification problem for each one of the single labels y_j which participates in the multi-label problem. In other words, one binary classification problem is created for each y_j , and each example in the training set is labeled as positive if y_j is an element of the respective multi-label example, negative otherwise. The final

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¹ List of abbreviations: LP, label powerset; BR, binary relevance; BR+, BRPlus; NU, no update; Stat, static order; Dyn, dynamic order; CC, classifiers chains; CLP, correlated label propagation; HOMER, hierarchy of multilabel classifiers; LC, label cardinality; LD, label density; NB, Naïve Bayes; CD, critical difference; MS, meta stacking.

labels for each multi-label example are determined by aggregating the classification results from all binary classifiers. Any supervised learning algorithm can be used as a base algorithm to generate these classifiers.

However, the *BR* approach presents two problems. Whenever the number of labels $|L|$ is large, the *BR* approach may experience drawbacks from the imbalanced data problem. The reason is that, in this case, it is more likely that at least for some binary classifiers, the number of negative examples that indicate that the specific label is not present, could be much larger than the number of positive examples. As a result, these binary classifiers might predict negative for all instances. Nevertheless, it should be noted that in some cases, learning algorithms perform well in imbalanced domains. As shown in Prati, Batista, and Monard (2004), it is not fair to directly correlate class imbalance to the loss of performance of learning algorithms. In fact, the problem is also related to the degree of overlapping among the classes. Thus, only imbalanced binary classifiers that do not predict the minority class well need to be treated separately in order to improve their performance. This can be done by using methods to deal with the problem of learning in the presence of class imbalance, as the ones described in Batista, Prati, and Monard (2004).

The other problem of the *BR* approach is due to its limitations to exploit dependency between the labels in the set of labels, and of its label-independence assumption. Nevertheless, the *BR* approach is simple to implement, intuitive and has low computation complexity compared to other methods, since it scales linearly with the size of the label set L .

The consensus view in the literature is that it is crucial to take into account label dependencies during the classification process. However, due to the exponentially expanding number of possible combinations of labels with increasing $|L|$, methods which model all label dependencies will have very high complexity. Thus, considering the low complexity of the *BR* method, it is worthwhile to find ways to overcome its disadvantages.

In this work, we describe a simple approach, named *BR+* (*BRplus*), which can be used to incorporate label dependency aiming to accurately predict label combinations. It should be observed that, unlike other approaches, *BR+* does not try to discover existing dependency among labels in advance before any classifier is induced. The general idea is to enable the classifiers to discover existing label dependency by themselves. This approach allows the user to specify different ways of considering the labels in order to explore label dependency. Three ways of considering the labels have already been implemented in *BR+*, which was initially presented in Cherman, Metz, and Monard (2010). This work expands our empirical evaluation of *BR+* by also comparing it with a previously proposed similar approach, known as Classifier Chains (Read, Pfahringer, Holmes, & Frank, 2009), as well as by using more multi-label datasets from different domains. Furthermore, for each dataset and base-algorithm, we show a simple simulation that can be carried out to calculate the best performance measures that could be obtained using the *BR+* approach. Results show that, for most of the datasets, the proposed *BR+* approach is able to improve the value of several of the multi-label data measures used in the evaluation.

This work is organized as follows: Section 2 briefly describes multi-label classification and the transformation problem Binary Relevance approach, as well as describes the approach proposed in this paper. Section 3 comments on solutions closely related to our proposal, as well as other solutions to discover label dependency. The results of the experimental evaluation of *BR+*, *CC* and *BR* on several multi-label datasets, as well as a simple simulation which can be carried out to estimate the best possible result that could be obtained by *BR+* on each dataset is presented in Sections 4 and 5. Section 6 concludes this work.

2. Material and methods

In this section, we briefly describe the background knowledge regarding the multi-label classification task and the proposed method, which makes use of the relationship among labels to improve the multi-label prediction quality. The definitions, as well as a well-known multi-label transformation method, called Binary Relevance, are detailed in Section 2.1, where some of the most frequently used evaluation measures and two dataset description measures are also presented. Furthermore, in Section 2.2 is presented a comprehensive explanation of the proposed *BR+* method, which is designed to take into account label dependency.

2.1. Multi-label classification and the *BR* approach

Let D be a training set with N examples $E_i = (x_i, Y_i)$, where $i = 1 \dots N$. Each instance E_i is associated with a feature vector $x = (x_{i1}, x_{i2}, \dots, x_{iM})$ and a subset of labels $Y_i \subseteq L$, where $L = \{y_j; j = 1 \dots q\}$ is the set of q possible labels. This representation is shown in Table 1. Considering this scenario, the multi-label classification task is to induce a classifier H that, given an unlabeled instance $E = (x, ?)$, is capable of accurately predicting its subset of labels Y , i.e., $H(E) \rightarrow Y$.

As stated earlier, the *BR* approach can be used to solve the multi-label classification problem. It consists of a problem transformation strategy that decomposes a multi-label classification problem into several distinct single-label binary classification problems, one for each label in set L of labels with $q = |L|$. The *BR* approach initially transforms the original training dataset into q binary datasets $D_{y_j}, j = 1 \dots q$, where each D_{y_j} contains all examples of the original dataset, but with a positive or negative label related to the single label y_j according to the true label subset associated with the example, i.e., positive if the label set contains label y_j and negative otherwise. The other labels ($y_k, k \neq j$) are not included in D_{y_j} . After the data is transformed, a set of q binary classifiers $H_j(E), j = 1 \dots q$ is constructed using the corresponding training dataset D_{y_j} . In other words, the *BR* approach initially constructs a set of q classifiers – Eq. (1):

$$H_{BR} = \{C_{y_j}((x, y_j)) \rightarrow y_j \in \{0, 1\} | y_j \in L : j = 1 \dots q\}. \quad (1)$$

To classify a new multi-label instance, the algorithm outputs the aggregation of the labels positively predicted by all the independent binary classifiers.

A relevant advantage of the *BR* approach is its low computational complexity compared with other multi-label methods. For a constant number of examples, *BR* scales linearly with size q of label set L . Thus, the *BR* approach is quite appropriate for not a very large q , which is the general case. Nevertheless, large number of labels can be found in various domains, and some divide-and-conquer methods have been proposed to organize labels into a tree-shaped hierarchy where it is possible to deal with a much smaller set of labels compared to q , as described in Section 4.

The performance of multi-label classifiers can be evaluated using different measures. Some of these measures are adaptations from the single-label classification evaluation measures, while

Table 1
Multi-label data.

	x				Y
E_1	x_{11}	x_{12}	...	x_{1M}	Y_1
E_2	x_{21}	x_{22}	...	x_{2M}	Y_2
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
E_N	x_{N1}	x_{N2}	...	x_{NM}	Y_N

others were specifically defined for multi-label tasks. A complete discussion on the performance measures for multi-label classification tasks is out of the scope of this paper, and can be found in Tsoumakas et al. (2009). In what follows, we briefly describe the measures used in this paper to compare our proposal with the *BR* and the *CC* approach. The measures are *Hamming-Loss*, *Accuracy*, *F-Measure* and *Subset-Accuracy*, defined by Eqs. (2)–(5) respectively:

$$\text{Hamming-Loss } (H, D) = \frac{1}{N} \sum_{i=1}^N \frac{|Y_i \Delta Z_i|}{|L|} \quad (2)$$

where Δ represents the symmetric difference between two sets, Y_i is the set of true labels and Z_i is the set of predicted labels. When considering the *Hamming-Loss* as the performance measure, the smaller the value, the better the algorithm performance is, with a lower bound equal to zero. For the next measures, greater values indicate better performance:

$$\text{Accuracy } (H, D) = \frac{1}{N} \sum_{i=1}^N \frac{|Y_i \cap Z_i|}{|Y_i \cup Z_i|}, \quad (3)$$

$$\text{F-Measure } (H, D) = \frac{1}{N} \sum_{i=1}^N \frac{2|Y_i \cap Z_i|}{|Z_i| + |Y_i|}, \quad (4)$$

$$\text{Subset-Accuracy } (H, D) = \frac{1}{N} \sum_{i=1}^N I(Z_i = Y_i), \quad (5)$$

where $I(\text{true}) = 1$ and $I(\text{false}) = 0$.

Subset-Accuracy is a very strict evaluation measure as it requires an exact match of the predicted and true set of labels.

Besides the performance measures, other measures to quantify the cardinality and the density of multi-label data are needed. The *label cardinality*, defined by Eq. (6), is the average number of labels associated with each example. The *label density*, defined by Eq. (7), is the normalized cardinality, i.e., the cardinality divided by the total number of single-labels in the set of possible labels L :

$$LC(D) = \frac{1}{|D|} \sum_{i=1}^{|D|} |Y_i|, \quad (6)$$

$$LD(D) = \frac{1}{|D|} \sum_{i=1}^{|D|} \frac{|Y_i|}{|L|}. \quad (7)$$

2.2. Proposed approach to incorporate label dependency

As stated earlier, *BR+* does not make any attempt to discover label dependency in advance. The main idea of *BR+* is to increment the feature space of the binary classifiers to let them discover existing label dependency by themselves.

In the training phase, *BR+* works in a similar manner to *BR*, i.e., q binary classifiers are generated, one for each label $y_j \in L$. However, there is a difference related to the q binary datasets used to generate the binary classifiers. In *BR+*, the feature space of these datasets is incremented with $q - 1$ features, which correspond to the other labels in the multi-label dataset.

In other words, each one of the D_{y_j} binary training datasets is augmented with ω_j binary features where $\omega_j = L - \{y_j\}$. Fig. 1, Training Phase, illustrates the binary datasets generated using a simple multi-label dataset with $L = \{y_1, y_2, y_3\}$. After this transformation, *BR+* constructs a set of q binary classifiers to classify new examples in this augmented feature space – Eq. (8):

$$H_{BR+} = \{C_{y_j}(x \cup \omega_j, y_j) \rightarrow y_j'' \in \{0, 1\} | \omega_j = L - \{y_j\}, y_j \in L: j = 1 \dots q\}. \quad (8)$$

The classification of a new example could be carried out in a similar way to the standard *BR* approach using these augmented

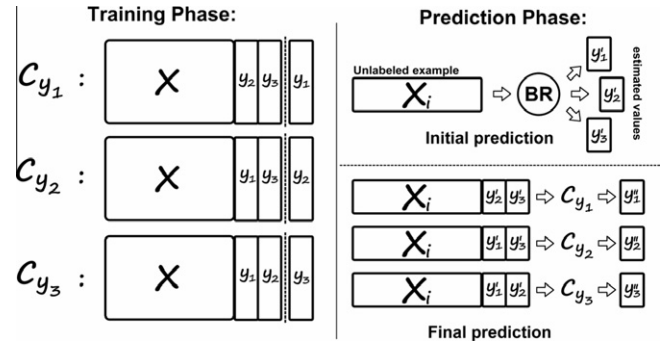


Fig. 1. Example of *BR+* with $L = \{y_1, y_2, y_3\}$.

binary classifiers. However, as each one of these classifiers has been generated using a feature space which is different to the original one, the unlabeled examples must also consider this augmented feature space. In other words, the feature space of the unlabeled examples must be augmented accordingly to the training set that generated the corresponding classifier.

At this point, *BR+* is faced with the following problem: The feature space of the unlabeled examples is different to the feature space of the training examples used to build the binary classifiers.

To solve this problem, it is necessary to apply the same transformation process used with the training examples for the unlabeled examples. To this end, the set of unlabeled examples is also replicated, resulting in q unlabeled set of examples with the corresponding feature space for each of the binary classifiers generated in the training phase. Furthermore, the value of these new features for the unlabeled examples is unknown, which leads to another problem:

How to estimate the values of the augmented features of unlabeled examples?

In this paper, the values of the new features are estimated by *BR+* in an intermediate step, where all the augmented feature values of unlabeled examples are initially predicted by *BR* (initial prediction in Fig. 1) considering the original training data. The values predicted by *BR* are used by *BR+* to complete the augmented feature space of the unlabeled examples. After this transformation, the unlabeled examples are classified by *BR+* in a similar way to the standard *BR* method, considering all the original features plus the augmented ones for each individual classifier. This prediction strategy, illustrated in Fig. 1 (final prediction), is called *NU* (no update), since no modification is made to the initial estimates of the augmented features during the prediction phase.

Furthermore, the *BR+* method can use a different strategy to predict the final set of labels for unlabeled examples, where the values of the initial prediction for the augmented features are updated with the prediction of the individual binary classifiers as the new predictions are obtained, considering a specific order of classification – Fig. 2.

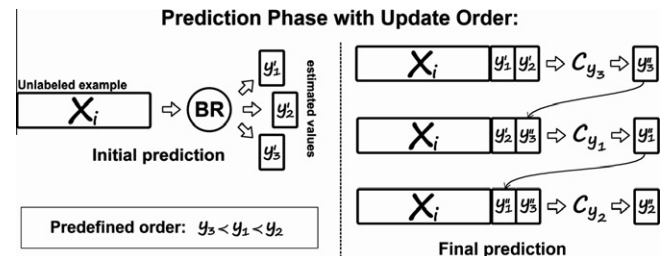


Fig. 2. Example of *BR+* prediction phase with update order. Predefined order: $y_3 < y_1 < y_2$.

For example, if a predefined order to predict the individual labels is considered, each of the new values y_j' can be used to update the previous value y_j' (initial prediction) of the corresponding augmented features of the unlabeled example.

To illustrate, consider as a predefined order the frequency of single labels in the training set, where the least frequent labels are predicted first, and consequently, updated first. We call this ordering, which does not depend on the classifiers, *Stat* (static order). Another possible order takes into account the confidence of the initial prediction for each independent single label given by *BR*, where labels predicted with less confidence are predicted and updated first in the final prediction phase of *BR+*. We call this ordering *Dyn* (dynamic order). Fig. 2 illustrates the update process considering a predefined order *Stat*.

Regarding *BR+* computational complexity, it can be observed that it simply duplicates *BR* complexity. Thus, *BR+* also scales linearly with the size q of the label set L . Therefore, it is appropriate for not a very large q .

3. Related work

Several attempts have been made to explore dependency between the labels in the set of labels using the *BR* approach. Most methods try to discover existing dependency among labels in advance before any classifier is induced, and few of them use only the classifiers to discover existing label dependency by themselves, as proposed in this work. One of these methods, called Meta Stacking (*MS*), is proposed in Godbole and Sarawagi (2004), where a stacked *BR* classification output indicating relationships between classes along with the full original feature space into a separate meta classifier is proposed, thereby creating a two-stage classification process. Although *MS* is similar to our approach, the authors use SVMs and propose a novel kernel for their method. We, on the other hand, are not restricted to any specific base-underlying algorithm. Moreover, they use all the outputs of the *BR* classifier to train the meta classifier in the second step. This can be a potential problem, since these new features are actually an estimate of the true label, made in the first *BR* classifier stage. Using these estimated values can lead the solution to an overfitted model. Our proposal, on the other hand, does not use the estimate for label y_j , i.e., y_j' , when training the individual final binary classifier in the *BR+* approach for that given (y_j) label, but only keeps the estimate for the other labels, consequently keeping the label correlation information in the set. Furthermore, in our work we include an empirical evaluation of different ways of considering the labels when building individual models using the *BR+* approach to effectively use the interrelationships between labels.

Another method, presented in Schapire and Singer (1999), maps confidence predictions of a single-label classifier to actual label subsets (observed in the training data) using Hamming distance. The subset with the shortest Hamming distance to the predictions is chosen as the predicted set. This method, called Subset Mapping (*SM*), can also be applied to the binary outputs of *BR*.

In Read et al. (2009), a similar approach as the one used by *BR+*, which also uses the concept of chains in the *BR* framework is proposed. This method, called Classifier Chains (*CC*), constructs a chain of classifiers and each classifier is responsible for learning and predicting the binary association of a specific label, given the feature space augmented by all prior binary classifiers predictions in the chain, i.e., the feature space of each link in the chain is extended with the 0–1 (present or not) label associations of all previous links. The *CC* method, which in this work is also compared to *BR+*, is illustrated in Fig. 3.

The standalone *CC* (Fig. 3) was compared with the *MS* and *SM* methods. Results show that *CC* improves over both methods *MS* and *SM*, and in many cases the difference is statistically significant.

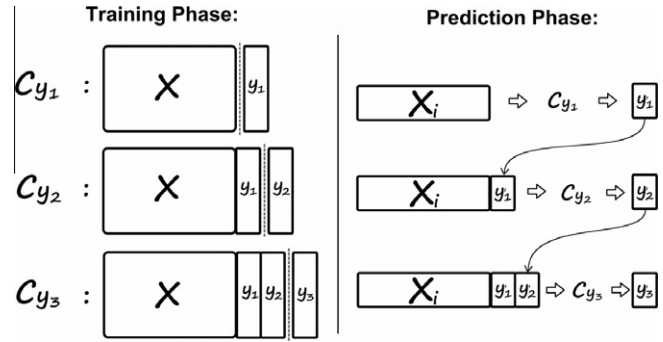


Fig. 3. Example of *CC* with predefined order: $y_1 < y_2 < y_3$.

Table 2

Description of the datasets used in the experiments.

Dataset	Domain	# Ex.	# Feat.	L	LC	LD	# Diff.
bibtex	Text	7395	1836	159	2.402	0.015	2856
corel5k	Images	5000	499	374	3.522	0.009	3175
emotions	Music	593	72	6	1.869	0.311	27
enron	Text	1702	1001	53	3.378	0.064	753
medical	Text	978	1449	45	1.245	0.028	94
scene	Images	2407	294	6	1.074	0.179	15
yeast	Biology	2417	103	14	4.237	0.303	198
slashdot-f	Text	3782	1079	22	1.180	0.041	156

Table 3

Results using SMO as base-algorithm.

SMO	BR	CC	BR+(NU)	BR+(Stat)	BR+(Dyn)
Hamming-Loss					
bibtex	0.02(0.00)	0.02(0.00)	0.02(0.00)	0.02(0.00)	0.02(0.00)
corel5k	0.01(0.00)	0.01(0.00)	0.02(0.00)	0.01(0.00)	0.01(0.00)
emotions	0.20(0.02)	0.23(0.03)	0.24(0.03)	0.24(0.04)	0.23(0.04)
enron	0.06(0.00)	0.06(0.00)	0.06(0.00)	0.06(0.00)	0.06(0.00)
medical	0.01(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)
scene	0.11(0.01)	0.10(0.01)	0.13(0.01)	0.11(0.01)	0.11(0.01)
slashdot	0.05(0.00)	0.05(0.01)	0.17(0.02)	0.05(0.00)	0.05(0.01)
yeast	0.20(0.01)	0.21(0.01)	0.22(0.01)	0.21(0.01)	0.21(0.01)
Subset-Accuracy					
bibtex	0.15(0.01)	0.16(0.01)	0.16(0.01)	0.16(0.01)	0.16(0.01)
corel5k	0.01(0.00)	0.02(0.01)	0.01(0.00)	0.01(0.00)	0.02(0.00)
emotions	0.27(0.06)	0.30(0.06)	0.22(0.06)	0.26(0.08)	0.25(0.08)
enron	0.11(0.03)	0.12(0.04)	0.11(0.04)	0.11(0.04)	0.12(0.03)
medical	0.66(0.06)	0.69(0.05)	0.69(0.05)	0.71(0.05)	0.69(0.05)
scene	0.51(0.03)	0.66(0.03)	0.54(0.03)	0.61(0.03)	0.61(0.04)
slashdot	0.37(0.05)	0.40(0.05)	0.38(0.05)	0.43(0.05)	0.43(0.05)
yeast	0.13(0.03)	0.20(0.04)	0.14(0.04)	0.19(0.04)	0.18(0.04)
F-Measure					
bibtex	0.40(0.01)	0.40(0.01)	0.41(0.01)	0.41(0.01)	0.41(0.01)
corel5k	0.11(0.01)	0.18(0.01)	0.23(0.00)	0.17(0.00)	0.21(0.00)
emotions	0.58(0.04)	0.60(0.05)	0.63(0.07)	0.59(0.08)	0.61(0.08)
enron	0.51(0.02)	0.51(0.02)	0.51(0.02)	0.51(0.02)	0.51(0.02)
medical	0.77(0.03)	0.79(0.03)	0.80(0.03)	0.80(0.03)	0.79(0.04)
scene	0.61(0.03)	0.72(0.03)	0.69(0.03)	0.70(0.03)	0.69(0.03)
slashdot	0.49(0.06)	0.52(0.05)	0.46(0.05)	0.50(0.05)	0.49(0.05)
yeast	0.60(0.02)	0.62(0.02)	0.60(0.02)	0.61(0.02)	0.61(0.02)
Accuracy					
bibtex	0.33(0.01)	0.33(0.01)	0.34(0.01)	0.34(0.01)	0.34(0.01)
corel5k	0.08(0.01)	0.13(0.01)	0.15(0.00)	0.12(0.00)	0.15(0.00)
emotions	0.50(0.04)	0.52(0.05)	0.53(0.07)	0.51(0.08)	0.52(0.08)
enron	0.40(0.03)	0.40(0.02)	0.40(0.03)	0.40(0.03)	0.40(0.02)
medical	0.74(0.04)	0.76(0.03)	0.77(0.04)	0.78(0.03)	0.76(0.04)
scene	0.58(0.03)	0.70(0.03)	0.65(0.03)	0.68(0.03)	0.67(0.03)
slashdot	0.46(0.05)	0.49(0.05)	0.43(0.05)	0.48(0.05)	0.48(0.05)
yeast	0.49(0.03)	0.52(0.03)	0.48(0.03)	0.51(0.03)	0.51(0.02)

Table 4
Results using J48 as base-algorithm.

J48	BR	CC	BR+(NU)	BR+(Stat)	BR+(Dyn)
<i>Hamming-Loss</i>					
<i>bibtex</i>	0.01(0.00)	0.01(0.00)	0.02(0.00)	0.02(0.00)	0.02(0.00)
<i>corel5k</i>	0.01(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)
<i>emotions</i>	0.24(0.04)	0.25(0.02)	0.27(0.03)	0.25(0.04)	0.25(0.04)
<i>enron</i>	0.05(0.00)	0.05(0.00)	0.06(0.00)	0.06(0.00)	0.06(0.00)
<i>medical</i>	0.01(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)
<i>scene</i>	0.14(0.01)	0.14(0.01)	0.19(0.02)	0.15(0.01)	0.15(0.01)
<i>slashdot</i>	0.04(0.00)	0.05(0.01)	0.25(0.02)	0.06(0.01)	0.06(0.01)
<i>yeast</i>	0.25(0.01)	0.27(0.01)	0.28(0.01)	0.29(0.01)	0.28(0.01)
<i>Subset-Accuracy</i>					
<i>bibtex</i>	0.14(0.01)	0.15(0.01)	0.16(0.01)	0.17(0.01)	0.17(0.01)
<i>corel5k</i>	0.00(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)
<i>emotions</i>	0.19(0.05)	0.19(0.05)	0.18(0.08)	0.25(0.08)	0.24(0.08)
<i>enron</i>	0.04(0.10)	0.12(0.04)	0.10(0.03)	0.13(0.03)	0.12(0.03)
<i>medical</i>	0.67(0.04)	0.69(0.04)	0.70(0.03)	0.71(0.04)	0.71(0.04)
<i>scene</i>	0.43(0.04)	0.53(0.03)	0.43(0.04)	0.51(0.05)	0.53(0.04)
<i>slashdot</i>	0.31(0.06)	0.34(0.06)	0.32(0.05)	0.34(0.05)	0.35(0.05)
<i>yeast</i>	0.05(0.02)	0.13(0.03)	0.06(0.02)	0.11(0.03)	0.10(0.03)
<i>F-Measure</i>					
<i>bibtex</i>	0.37(0.01)	0.36(0.02)	0.40(0.02)	0.40(0.02)	0.41(0.02)
<i>corel5k</i>	0.09(0.01)	0.07(0.01)	0.13(0.01)	0.12(0.01)	0.13(0.01)
<i>emotions</i>	0.53(0.08)	0.53(0.05)	0.53(0.06)	0.55(0.06)	0.55(0.08)
<i>enron</i>	0.50(0.03)	0.51(0.03)	0.52(0.03)	0.52(0.03)	0.51(0.03)
<i>medical</i>	0.79(0.04)	0.81(0.04)	0.82(0.04)	0.83(0.04)	0.82(0.04)
<i>scene</i>	0.55(0.03)	0.60(0.03)	0.57(0.04)	0.57(0.04)	0.58(0.04)
<i>slashdot</i>	0.37(0.06)	0.40(0.06)	0.45(0.05)	0.41(0.06)	0.42(0.06)
<i>yeast</i>	0.55(0.02)	0.52(0.03)	0.54(0.02)	0.53(0.03)	0.58(0.02)
<i>Accuracy</i>					
<i>bibtex</i>	0.31(0.01)	0.30(0.02)	0.33(0.02)	0.34(0.02)	0.34(0.02)
<i>corel5k</i>	0.06(0.00)	0.05(0.01)	0.09(0.01)	0.08(0.01)	0.09(0.01)
<i>emotions</i>	0.44(0.07)	0.45(0.05)	0.44(0.06)	0.47(0.06)	0.47(0.07)
<i>enron</i>	0.40(0.03)	0.41(0.03)	0.40(0.03)	0.42(0.03)	0.41(0.03)
<i>medical</i>	0.76(0.03)	0.78(0.04)	0.79(0.04)	0.80(0.04)	0.80(0.04)
<i>scene</i>	0.52(0.03)	0.59(0.03)	0.53(0.04)	0.55(0.04)	0.57(0.04)
<i>slashdot</i>	0.36(0.06)	0.38(0.06)	0.40(0.06)	0.39(0.06)	0.41(0.06)
<i>yeast</i>	0.42(0.02)	0.42(0.02)	0.41(0.02)	0.42(0.03)	0.46(0.02)

Regarding the chain order, several possible heuristics can be used, although the effect of different orders on the prediction performance of the algorithm was not studied in depth. The authors solved this issue by using an ensemble framework with a different random chain ordering per iteration. The predictions are summed by label so that each label receives a number of votes. A threshold is used to select the most popular labels, which form the final predicted multi-label set.

Various methods which try to discover existing dependency among labels in advance have been proposed, although in most of them the efforts in exploiting label correlation information are limited to pair wise correlation of class labels. One attempt to overcome these issues is proposed in (Kang, Jin, & Sukthankar, 2006), where a novel framework for multi-label learning is presented. This framework, called Correlated Label Propagation (CLP), explicitly models interactions between labels and simultaneously co-propagates multiple labels from the training examples to test examples using their similarities.

However, this method needs the specification of a set of weights that determines the relative order of the labels to be propagated. The order can be selected using some kind of heuristic function such as class frequency or the reverse of the class frequency. The authors of CLP have chosen the inverse of the class frequency since it allows the rare classes to determine their confidence scores before the popular classes. Nevertheless, there is still a need for an extensive evaluation of different selection order heuristics.

Table 5
Results using NB as base-algorithm.

NB	BR	CC	BR+(NU)	BR+(Stat)	BR+(Dyn)
<i>Hamming-Loss</i>					
<i>bibtex</i>	0.09(0.00)	0.14(0.01)	0.15(0.01)	0.21(0.01)	0.17(0.01)
<i>corel5k</i>	0.01(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)
<i>emotions</i>	0.25(0.04)	0.25(0.04)	0.21(0.04)	0.20(0.04)	0.21(0.04)
<i>enron</i>	0.20(0.01)	0.20(0.01)	0.20(0.01)	0.20(0.01)	0.20(0.01)
<i>medical</i>	0.02(0.00)	0.02(0.00)	0.03(0.00)	0.03(0.00)	0.03(0.00)
<i>scene</i>	0.24(0.01)	0.24(0.01)	0.16(0.01)	0.16(0.01)	0.16(0.01)
<i>slashdot</i>	0.07(0.01)	0.06(0.01)	0.05(0.01)	0.05(0.01)	0.05(0.01)
<i>yeast</i>	0.29(0.02)	0.29(0.02)	0.28(0.02)	0.28(0.02)	0.29(0.02)
<i>Subset-Accuracy</i>					
<i>bibtex</i>	0.06(0.01)	0.06(0.01)	0.07(0.01)	0.07(0.01)	0.07(0.01)
<i>corel5k</i>	0.00(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)	0.01(0.00)
<i>emotions</i>	0.21(0.07)	0.23(0.07)	0.30(0.07)	0.33(0.08)	0.33(0.08)
<i>enron</i>	0.00(0.00)	0.00(0.00)	0.00(0.00)	0.00(0.00)	0.00(0.00)
<i>medical</i>	0.19(0.04)	0.19(0.04)	0.33(0.07)	0.33(0.07)	0.33(0.07)
<i>scene</i>	0.18(0.04)	0.18(0.04)	0.32(0.04)	0.32(0.04)	0.32(0.04)
<i>slashdot</i>	0.23(0.04)	0.28(0.06)	0.44(0.06)	0.44(0.05)	0.44(0.05)
<i>yeast</i>	0.10(0.02)	0.11(0.03)	0.11(0.03)	0.12(0.03)	0.12(0.03)
<i>F-Measure</i>					
<i>bibtex</i>	0.26(0.00)	0.23(0.01)	0.23(0.01)	0.21(0.01)	0.22(0.01)
<i>corel5k</i>	0.22(0.01)	0.21(0.01)	0.23(0.01)	0.24(0.01)	0.24(0.01)
<i>emotions</i>	0.63(0.03)	0.63(0.03)	0.65(0.06)	0.66(0.06)	0.66(0.06)
<i>enron</i>	0.32(0.01)	0.32(0.01)	0.32(0.02)	0.32(0.02)	0.32(0.02)
<i>medical</i>	0.30(0.04)	0.30(0.04)	0.46(0.06)	0.46(0.06)	0.46(0.06)
<i>scene</i>	0.56(0.03)	0.56(0.03)	0.69(0.02)	0.68(0.02)	0.68(0.02)
<i>slashdot</i>	0.44(0.03)	0.48(0.05)	0.55(0.06)	0.55(0.05)	0.56(0.05)
<i>yeast</i>	0.55(0.02)	0.55(0.03)	0.55(0.03)	0.55(0.03)	0.55(0.03)
<i>Accuracy</i>					
<i>bibtex</i>	0.19(0.00)	0.17(0.01)	0.17(0.01)	0.16(0.01)	0.16(0.01)
<i>corel5k</i>	0.15(0.00)	0.15(0.01)	0.16(0.01)	0.17(0.01)	0.17(0.01)
<i>emotions</i>	0.53(0.04)	0.53(0.04)	0.56(0.06)	0.58(0.06)	0.58(0.06)
<i>enron</i>	0.21(0.01)	0.21(0.01)	0.21(0.01)	0.21(0.01)	0.21(0.01)
<i>medical</i>	0.27(0.04)	0.27(0.04)	0.42(0.06)	0.42(0.06)	0.42(0.06)
<i>scene</i>	0.45(0.03)	0.45(0.03)	0.59(0.02)	0.59(0.02)	0.59(0.02)
<i>slashdot</i>	0.38(0.04)	0.43(0.05)	0.52(0.06)	0.52(0.05)	0.53(0.05)
<i>yeast</i>	0.43(0.03)	0.44(0.03)	0.43(0.03)	0.44(0.03)	0.44(0.03)

In Ji, Tang, Yu, and Ye (2008) a general framework for extracting shared structures in multi-label classification is proposed. The authors advocate that multiple labels share the same input space, and the semantics conveyed by different labels are usually correlated. Taking this into account, a low-dimensional subspace is assumed to be shared among multiple labels. Then, a binary classifier is constructed to discriminate each label from the rest of them using the original data space and the representation in the shared subspace. This is computationally expensive, despite an approximation algorithm used in their experimental setup where they randomly select a fraction of the data points for training.

An algorithm that works on a hypergraph constructed to exploit the correlation among the different labels is proposed in Sun, Ji, and Ye (2008). A weak point of this method is the computational complexity of the hypergraph, although the authors also presented an approximate formulation for the algorithm that should reduce the computational complexity. However, this approximation is not sufficient to scale up the method to large datasets with a large number of distinct labels.

Another algorithm where label dependency is analyzed first is proposed in Tenenboim, Shapira, and Rokach (2009). After analyzing label dependency, the whole set of labels is divided into several mutually exclusive subsets, and finally a classification algorithm incorporating dependencies among labels within each subset can be applied.

Although label dependency is a central problem in multi-label classification, the high dimensionality of the label space

Table 6
Number of best results for each base-algorithm/number of overall best results.

	SMO			J48			NB		
	Subset-Accuracy	F-Measure	Accuracy	Subset-Accuracy	F-Measure	Acc.	Subset-Accuracy	F-Measure	Accuracy
BR	0/0	1/0	1/0	0/0	0/0	0/0	1/0	3/0	2/0
CC	6/3	4/2	4/2	3/0	1/0	1/0	2/0	2/0	2/0
BR+(NU)	1/0	5/1	4/1	1/0	3/1	1/0	6/1	4/0	3/0
BR+(Stat)	3/1	3/1	3/1	5/3	3/2	4/3	8/2	5/2	6/2
BR+(Dyn)	4/1	2/1	3/1	5/2	4/1	6/2	8/3	6/3	7/3

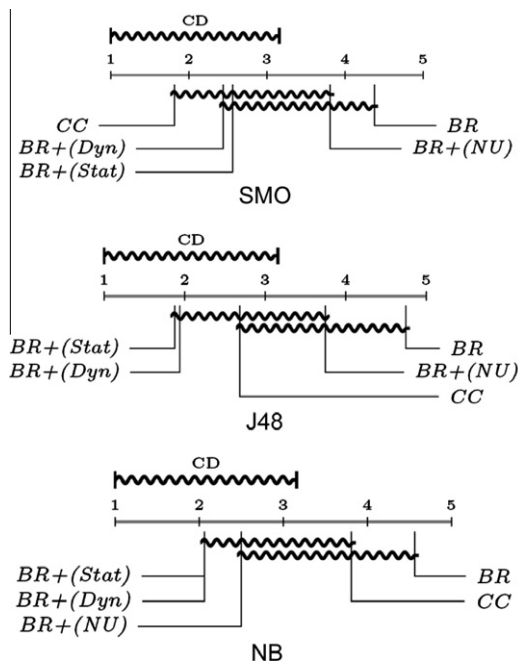


Fig. 4. Visualization of the Nemenyi's post hoc test for Subset-Accuracy. Groups of methods which are not significantly different at $p < 0.05$ are connected.

may also challenge the learning algorithms. In Tsoumakas, Katakis, and Vlahavas (2008), in order to scale up multi-label algorithms to domains with large number of labels *HOMER* (Hierarchy Of Multilabel classifERs) is proposed. The algorithm follows the divide-and-conquer strategy. The main idea is to transform a multi-label classification task with a large set of labels into a tree-shaped hierarchy of simpler tasks, each of them dealing with a small number of labels. These labels are distributed into disjoint subsets so that similar labels are placed together and dissimilar labels are kept apart. The authors use a new balanced clustering algorithm, called balanced k-means, to distribute the subsets of labels in the different nodes of the hierarchy.

Another method to solve the problem of large number of labels is proposed in Zhang et al. (2010). The method is based on Random Decision Tree for multi-label classification and it is called multi-label RDT. Compared with *HOMER*, multi-label RDT runs 2–3 orders of magnitude faster in training process and achieves some improvement on accuracy.

4. Results

In this section, the performance of *BR+*, *BR* and *CC* are analyzed considering the three different ways to classify a new multi-label instance already implemented in *BR+*, the no update *NU*, the static (*Stat*) and the dynamic (*Dyn*) update methods. The experiments were repeated using three different base single-label classifiers.

4.1. Datasets

To make this comparison, we selected eight multi-label datasets from different domains, which can be found in the Mulan website.² Table 2 summarizes the datasets characteristics. It shows, for each dataset, the number of examples (# ex.), the number of features (# feat.), the number of possible labels ($|L|$), the label cardinality (*LC*) and label density (*LD*) of the multi-label data, as well as the number of different sets of labels (# diff.).

4.2. Experimental setup

BR+ was implemented using Mulan,³ a package of Java classes for multi-label classification based on Weka.⁴ Furthermore, we used the *CC* version implemented by its authors in the Weka framework, which was also incorporated into the Mulan framework, where *BR* is already implemented. The experiments were carried out using three different base single-label classifiers from Weka. Two are discriminative, J48, a decision tree learning algorithm and the support vector machine SMO learning algorithm. The third is the generative Naïve Bayes (NB) learning algorithm. The three algorithms were executed using default parameters.

All the reported results were obtained using 10-fold cross validation with paired folds, i.e., the same training and testing partitions were used to obtain the average of the four measures used to compare *BR*, *CC* and *BR+*.

4.3. Results

Tables 3–5 show, respectively, the average of the four measure values, for each base-classifier, i.e. SMO, J48 and NB, used in the experiments. The numbers in brackets correspond to the standard deviation. For the sake of visualization, the best measure values for each base-classifier are shown in bold. The underlined values show the best measure values taking into account the results of the three base-classifiers. In other words, the overall best results obtained when comparing the results of the SMO, J48 and NB classifiers.

5. Discussion

Results show that, in general, the Hamming-Loss measure does not improve for *BR+*. Observe that in most cases where *BR+* achieved good results, there is a draw with the other algorithms. However, the other measure values are better for *BR+* most of the time, and in some cases they are significantly better than *BR*.

Comparing the results obtained using the three different base single-label classifiers in each dataset, it is possible to observe that, as expected, they have a strong influence. It is worth noting the very poor results obtained with dataset corel5k by the three base

² <http://mulan.sourceforge.net/datasets.html>.

³ <http://mulan.sourceforge.net/>.

⁴ <http://www.cs.waikato.ac.nz/ml/weka/>.

single-label classifiers. In this dataset $|L| = 374$ – Table 2, and the majority of the 374 binary classifiers constructed in the first step of $BR+$ or the CC have very low precision, which justifies the poor results obtained using this dataset. To sum up, Table 6 shows the number of times that the best values of Accuracy, F-Measure and Subset-Accuracy were achieved for each base single-label classifier, as well as the number of times that these values were the best among all of them. Note that although draws are included in these numbers, BR did not achieve any overall best result.

As can be observed, the performance of CC is better when using SMO, although it degrades when using J48 and NB, where no overall best result was obtained by CC . This is not the case for $BR+$, which shows overall best results, except for J48 and the NU update method, in all the other cases.

In order to analyze whether there is a difference among the methods, for each one of the four measures and the three single-label base-classifier, we ran the Friedman's test with the null-hypotheses that the performance of all methods are comparable considering all results. In only one of the 12 cases the null-hypothesis was not rejected. Specifically, it was not rejected for the Hamming-Loss measure with Naïve Bayes as a single-label base-classifier – Table 5.

When the null-hypothesis is rejected by the Friedman's test, at 95% of confidence level, we can proceed with a post hoc test to detect which differences among the methods are significant. We ran the Nemenyi's multiple comparison test, comparing all algorithms, where the Nemenyi test points out whether there is a significant difference between the methods involved in the experimental evaluation, whenever their average rank differs by at least the critical value.

It has been observed that sometimes the Friedman's test reports a significant difference, but the post hoc test fails to detect it. This is due to the lower power of the post hoc test. In this case, the only conclusion that can be drawn is that some algorithms do differ significantly (Demšar, 2006). In our case, the Nemenyi's test failed to detect significant difference for SMO and NB, except for the Subset-

Accuracy measure. The results of the Nemenyi's test can be represented in a simple diagram illustrated in the following figures, where the average ranks of the methods is plotted on the X-axis (lower average rank is better). The critical difference (CD) is shown above the X-axis, and the secondary lines beneath the X-axis connect groups which are not significantly different. Fig. 4 shows the Subset-Accuracy results for the three single-label base-classifier used in the experiments.

As seen in Fig. 4, for SMO, CC performs best, followed by $BR+(Dyn)$, $BR+(Stat)$, $BR+(NU)$ and BR in that order. Although the Nemenyi's test points out that CC outperforms all the other algorithms, the only significant difference is between CC and BR . For J48, $BR+(Stat)$ performs best, followed by $BR+(Dyn)$, CC , $BR+(NU)$ and BR in that order, where the only significant difference is observed when comparing methods $BR+(Stat)$ and $BR+(Dyn)$ with BR . For NB, results are similar as for J48 regarding the significant difference, although in this case $BR+(NU)$ performs better than CC . In general, it can be observed that $BR+$ as well as CC outperform BR . Moreover, considering the average ranks, $BR+$ shows to be a good competitor of CC . It follows the order of the average ranks obtained in the experiments:

Subset-Accuracy:

SMO: $CC \prec BR + (Dyn) \prec BR + (Stat) \prec BR + (NU) \prec BR$

J48: $BR + (Stat) \prec BR + (Dyn) \prec CC \prec BR + (NU) \prec BR$

NB: $BR + (Stat) \prec BR + (Dyn) \prec BR + (NU) \prec CC \prec BR$

F-Measure:

SMO: $CC \prec BR + (Stat) \prec BR + (NU) \prec BR + (Dyn) \prec BR$

J48: $BR + (Dyn) \prec BR + (NU) \prec BR + (Stat) \prec CC \prec BR$

NB: $BR + (Dyn) \prec BR + (NU) \prec BR + (Stat) \prec BR \prec CC$

Accuracy:

SMO: $CC \prec BR + (Dyn) \prec BR + (Stat) \prec BR + (NU) \prec BR$

J48: $BR + (Dyn) \prec BR + (Stat) \prec CC$ and $BR + (NU) \prec BR$

NB: $BR + (Dyn) \prec BR + (Stat) \prec BR + (NU) \prec CC \prec BR$

Table 7
Simulated $BR+$ vs BR .

	F-Measure		Hamming-Loss		Accuracy		Subset-Accuracy	
	Simulated	BR	Simulated	BR	Simulated	BR	Simulated	BR
SMO								
bibtex	0.45(0.01)	0.40(0.01)	0.01(0.00)	0.02(0.00)	0.37(0.01)	0.33(0.01)	0.17(0.01)	0.15(0.01)
corel5k	0.48(0.01)	0.11(0.01)	0.01(0.00)	0.01(0.00)	0.37(0.01)	0.08(0.01)	0.06(0.01)	0.01(0.01)
emotions	0.70(0.06)	0.58(0.04)	0.17(0.03)	0.20(0.02)	0.60(0.06)	0.50(0.04)	0.30(0.08)	0.27(0.04)
enron	0.61(0.01)	0.51(0.02)	0.05(0.00)	0.06(0.00)	0.49(0.02)	0.40(0.03)	0.15(0.04)	0.11(0.03)
medical	0.85(0.04)	0.77(0.03)	0.01(0.00)	0.01(0.00)	0.82(0.04)	0.74(0.04)	0.73(0.06)	0.66(0.03)
scene	0.84(0.03)	0.61(0.03)	0.07(0.01)	0.11(0.01)	0.81(0.03)	0.58(0.03)	0.73(0.03)	0.51(0.03)
slashdot	0.84(0.03)	0.49(0.06)	0.02(0.00)	0.05(0.00)	0.83(0.03)	0.46(0.05)	0.79(0.03)	0.37(0.05)
yeast	0.94(0.01)	0.60(0.02)	0.04(0.00)	0.20(0.01)	0.90(0.01)	0.49(0.03)	0.69(0.04)	0.13(0.02)
J48								
bibtex	0.45(0.02)	0.37(0.01)	0.01(0.00)	0.01(0.00)	0.38(0.02)	0.31(0.01)	0.18(0.01)	0.14(0.01)
corel5k	0.43(0.01)	0.09(0.01)	0.01(0.00)	0.01(0.00)	0.33(0.01)	0.06(0.00)	0.06(0.01)	0.00(0.00)
emotions	0.72(0.04)	0.53(0.08)	0.16(0.02)	0.24(0.04)	0.64(0.04)	0.44(0.07)	0.37(0.07)	0.19(0.05)
enron	0.63(0.02)	0.50(0.03)	0.04(0.00)	0.05(0.00)	0.52(0.02)	0.40(0.03)	0.17(0.03)	0.10(0.04)
medical	0.85(0.03)	0.79(0.04)	0.01(0.00)	0.01(0.00)	0.82(0.03)	0.76(0.03)	0.73(0.03)	0.67(0.04)
scene	0.85(0.03)	0.55(0.03)	0.06(0.01)	0.14(0.01)	0.83(0.03)	0.52(0.03)	0.77(0.04)	0.43(0.04)
slashdot	0.84(0.03)	0.37(0.06)	0.02(0.00)	0.04(0.00)	0.81(0.04)	0.36(0.06)	0.74(0.04)	0.31(0.06)
yeast	0.94(0.01)	0.55(0.02)	0.04(0.01)	0.25(0.01)	0.90(0.02)	0.42(0.02)	0.72(0.04)	0.05(0.02)
NB								
bibtex	0.28(0.01)	0.26(0.00)	0.08(0.00)	0.09(0.00)	0.20(0.01)	0.19(0.00)	0.07(0.01)	0.06(0.01)
corel5k	0.43(0.01)	0.22(0.01)	0.01(0.00)	0.01(0.01)	0.32(0.01)	0.15(0.00)	0.03(0.01)	0.00(0.00)
emotions	0.72(0.05)	0.63(0.03)	0.18(0.03)	0.25(0.03)	0.63(0.06)	0.53(0.04)	0.35(0.09)	0.21(0.07)
enron	0.33(0.01)	0.32(0.01)	0.19(0.01)	0.20(0.01)	0.22(0.01)	0.21(0.01)	0.00(0.00)	0.00(0.00)
medical	0.46(0.06)	0.30(0.04)	0.03(0.00)	0.02(0.04)	0.43(0.06)	0.27(0.04)	0.33(0.07)	0.19(0.04)
scene	0.69(0.02)	0.56(0.03)	0.16(0.01)	0.24(0.03)	0.59(0.02)	0.45(0.03)	0.32(0.04)	0.18(0.04)
slashdot	0.64(0.04)	0.44(0.03)	0.04(0.00)	0.07(0.01)	0.61(0.04)	0.38(0.03)	0.52(0.04)	0.23(0.04)
yeast	0.79(0.02)	0.55(0.02)	0.12(0.01)	0.29(0.02)	0.69(0.02)	0.43(0.03)	0.25(0.03)	0.10(0.02)

Furthermore, there is a simple way to find out the best results that *BR+* can achieve compared with *BR*, which does not consider label dependency. The best results will be obtained by *BR+* whenever the initial estimate of each y_j label, given by y'_j , is perfect. Using this idea, *BR+* was executed in a simulated mode (denoted as *Simulated* in Table 7, where the numbers in brackets correspond to standard deviation) using $y'_j = y_j$ in the second step of *BR+*.

As can be observed, except for emotions, scene, slashdot-f and yeast datasets, there is little room to improve the Hamming-Loss measure by using *BR+*. However, this is not the case with F-Measure and Accuracy. In these cases, not considering the dataset bibtex, there is room for improvement, with special attention to dataset corel5k. Although there is room for Subset-Accuracy improvements, it can be observed that the value of this measure is low for the first four datasets although it could be acceptable for the emotions dataset.

In fact, given a multi-label dataset it is worthwhile executing *BR+* in the simulated mode and compare the results obtained by different measures using different base single-label classifiers. Based on these results, it is possible to focus on the ones for which better results could be expected, and further concentrate on improving the results obtained by these specific classifiers, thus simplifying the experimental search for the best single-label classifier.

6. Conclusions

This paper presented *BR+*, a simple approach for multi-label classification, which is based on the Binary Relevance approach. Its main objective consists of exploring label dependency by only using the binary classifiers to discover and accurately predict label combinations. To this end, three different ways to classify a new multi-label instance were proposed. Using several multi-label datasets from different domains and various evaluation measures, we carried out an empirical evaluation to compare both approaches and to verify if the proposed approach is able to predict label combinations more accurately than *BR*. We also considered a similar method, the Chain Classifiers, which is considered a very good method, in our evaluation.

Initial results show the potential of the proposed approach to improve the multi-label classification performance in datasets which do not have a very high dimensionality of the label space. Although difference statistically significant of *BR+* and *CC* was not found, results show that *BR+* improves over *CC* most of the time. For datasets with high label space dimensionality, an upgrading algorithm such as *HOMER* (Tsoumakas et al., 2008) should be used first. Furthermore, it is shown that a simple simulation, which is independent of the order used to predict the individual labels, can be carried out to determine the best possible results that *BR+* can achieve in a dataset, enabling the user to narrow the search for the best base-classifier for any specific dataset using *BR+*. However, this feature cannot be implemented in *CC*, unless the chain order is known in advance. Nevertheless, more extensive experiments should be done in the future in order to validate these results.

The performance of multi-label classification depends on many factors, which are hard to isolate. As future work, we plan to investigate further label dependency looking at the problem from a statistical perspective, as suggested in Dembczynski, Waegeman, Cheng, and Hüllermeier (2010), in order to theoretically understand the reasons for improvements of the different measures used in multi-label classification.

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